ANALYTICAL APPROACH TO MARKETING DECISIONS

MARKETING CAMPAIGN ANALYSIS

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1. Problem Statement

The main goal of this project is to better understand different types of customers so the company can market more effectively. Rather than promoting all products to everyone, data will be used to uncover patterns in customer behaviour. These insights will help identify distinct customer groups based on spending habits, lifestyle, and demographics. As a result, the business can design personalized campaigns that build loyalty and drive higher sales.

2. Dataset Dictionary

The Customer Personality Analysis dataset from Kaggle is a marketing dataset based on past campaigns of a retail company. It contains 2,240 customer records with 29 features, offering rich information on customer demographics and lifestyles. The dataset also captures purchasing behaviour and responses to previous marketing campaigns. It provides valuable insights into how customers interact with different products and promotions. This data is widely used for customer segmentation and predictive modeling. Overall, it helps businesses better understand their customers and design more effective marketing strategies.
The description of each variable is as follows:

- **ID:** Unique customer identifier.
- **Demographics:** Year_Birth, Education, Marital_Status, and Income.
- **Household:** Kidhome and Teenhome (counts of children and teenagers).
- **Dt_Customer:** Date the customer enrolled.
- **Recency:** Days since the last purchase.
- **Spending (Mnt...):** Amount spent in the last 2 years on Wines, Fruits, Meat, Fish, Sweets, and Gold.
- **Purchase Channels (Num...):** Count of purchases made via Web, Catalog, and Store.
- **Engagement:** NumDealsPurchases (purchases with discount) and NumWebVisitsMonth.
- **Campaign History:** AcceptedCmp1 through 5 (previous campaigns) and Response (most recent campaign).
- **Other:** Complain (complaint history) and internal variables Z_CostContact /Z_Revenue.

3. Exploratory Data Analysis (EDA)

3.1.Data Cleaning and Preparation

We began by importing the marketing_campaign.csv file into R and explored the dataset using the head(), tail(), and str() functions. Each row represents a single customer, with a mix of numeric variables like income, recency, and product spending.The dataset also includes categorical variables such as education level, marital status, response, and complaints. The Dt_Customer column was converted into a proper date format to accurately reflect customer joining dates. Additionally, key categorical variables were transformed into factors to support effective grouping and modelling.
Then we cleaned the dataset by:

- Removing rows with missing Income values because Income is a significant predictor.

- Filtering out customers with Income above 200,000 to remove extreme outliers.
- Eliminating duplicate records to avoid counting the same customer twice.
- Dropping the constant columns Z_CostContact and Z_Revenue, which don't add information.
- Cleaning text in Marital_Status (removing extra spaces, making case consistent).
- Using complete cases () to remove any rows with missing data.

Outlier Summary: Outliers were limited, raw dataset (2,241 rows) vs cleaned dataset (2,213 rows) shows only 29 rows removed (1.1% data loss). This minimal data loss improves model stability while retaining nearly all customer information.
After cleaning, we had 2,212 valid customers and 28 useful variables, ready for analysis and modelling.

3.2.Descriptive Statistics and Target Balance

We used summary () and skim () functions in R to obtain descriptive statistics for all variables.
Key Numeric of Income Variable:

- Minimum: ~$1,730
- Median: ~$51,373
- Mean: ~$51,970
- Maximum: ~$162,397

Most customers are in the mid-income range ($30,000-$70,000), with a smaller group of high-income customers.
Education Distribution:

- Graduation: 1,115 customers (50%)
- PhD: 481 customers (22%)
- Master: 365 customers (16%)
- 2nd Cycle: 200 customers (9%)
- Basic: 54 customers (2%)

3.3.Response Variable (Target Variable) Distribution

The response variable shows significant class imbalance: 85% no-response, 15% positive response. This imbalance requires evaluation using recall, precision, and F1-score rather than accuracy alone. A naive model always predicting "No" would achieve 85% accuracy but zero business value. This distribution is critical for model evaluation and requires careful consideration of appropriate performance metrics.
See Appendix B.1 for detailed statistics

3.4.Income Distribution and Transformation

Raw Income Distribution:
Income is strongly right-skewed (median $51,373). Most customers earn $40,000-$60,000, while a few exceed $150,000. This right skew justifies our outlier filtering at $200,000.
Log transformation explores and reduces skewness, but the original Income retained for model interpretability.

See Appendix B.2 for income distribution histograms

3.5. Relationship Between Education and Income

Higher education is associated with higher income:

- **PhD:** Median income ~$61,500 (highest earners)
- **Master:** Median income ~$57,800
- **Graduation:** Median income ~$51,200 (largest group, 50% of customers)
- **2nd Cycle:** Median income ~$40,100
- **Basic:** Median income ~$27,500 (smallest group, 2%)

Education is a strong predictor of income and helps identify higher-value customer segments.
See Appendix B.3 for education-income boxplot

3.6. Income and Product Spending

Income strongly correlates with premium product spending**:**

- **Income vs Wine Spending:** Correlation = 0.71 (strong positive)
- **Income vs Meat Spending:** Correlation = 0.68 (strong positive)

Higher-income customers spend significantly more on premium product categories. No multicollinearity issues were detected.
See Appendix B.4 for scatterplots and correlation matrix

3.7. Income and Campaign Response

Customers who responded have higher median income:

- Responders (Response=1): ~$62,500 median
- Non-responders (Response=0): ~$50,000 median
- Difference: ~$12,500

However, considerable overlap shows that income alone cannot perfectly predict response. Additional variables are needed for accurate prediction.
See Appendix B.5 for response-income boxplot

3.8. Key Findings and Conclusions

- Income Distribution: Strongly right-skewed with a median of $51,373
- Response Imbalance: 85% no-response, 15% positive response requires careful metric selection
- Education Effect: Higher education (Graduation, PhD, Master) strongly associated with higher incomes
- Spending Patterns: Income positively correlates with premium product spending (wine, meat)
- Data Quality: Successfully cleaned with only 1.1% data loss (25 rows), no missing values, ready for modelling
- Model Readiness: Low multicollinearity and proper data types ensure stable model building

The cleaned dataset is now ready for predictive modelling to identify customer segments and develop targeted marketing strategies.

4. Model Building

4.1.Decision Tree

A decision tree model was developed to predict customer response to a marketing campaign (Response = 1 for responders, 0 otherwise). The primary objective was interpretability understanding, which customers respond and in what sequence of decisions rather than maximizing predictive accuracy. The model was implemented in R using the *rpart* package and trained on the same cleaned dataset and 70/30 train–test split used for logistic regression to allow direct comparison. Customer ID and enrollment date were excluded, and the response variable was converted to a factor for classification. To reduce overfitting, the tree was trained with growth constraints (minsplit = 20, cp = 0.01) and then pruned using cross-validated error, resulting in a simpler and more interpretable final tree. The tree classifies customers through a sequence of binary splits. The most important split is based on previous campaign acceptance, particularly *AcceptedCmp5*, indicating that past campaign response is the strongest predictor of future response. Customers who accepted prior campaigns are much more likely to respond again, while those who did not are further evaluated using tenure and online engagement. For customers with no recent campaign acceptance, longer tenure, and higher website activity increase the likelihood of response, whereas newer and less engaged customers are typically classified as non-responders. This demonstrates the model's ability to distinguish inactive customers from those who remain potentially responsive.

Variable importance was assessed using 5-fold cross-validation with the *caret* package. The most influential predictors include previous campaign acceptance (Campaigns 3, 1 and 5), total customer spending, tenure and recency, and website activity which is consistent with the logistic regression findings. Model performance was evaluated using confusion matrices and ROC curves, with the decision tree achieving a test AUC of approximately 0.80, indicating reasonable discriminatory power. Although slightly less accurate than logistic regression, the decision tree offers superior interpretability and yields actionable marketing rules, such as prioritizing customers who accepted previous campaigns and targeting loyal, engaged customers even if they ignored recent offers. Overall, the decision tree reinforces the key drivers identified by logistic regression while providing a transparent framework for marketing decisions.

4.2.Logistic Regression

A logistic regression model was developed to predict customer response in a marketing campaign dataset. Logistic regression was selected because the dependent variable is binary and the model produces probability-based predictions, which are essential for threshold selection and profitability analysis. A stepwise regression approach was applied to retain statistically significant predictors while reducing model complexity. The dataset was split into 70% training data and 30% test data to evaluate out-of-sample performance.

The fitted model generated predicted probabilities for both training and test observations. These probabilities were exported to an Excel file named MC_modellingfile.csv, which contains customer IDs, actual responses, dataset indicators (train/test), predicted probabilities, and associated profit calculations. Using this modeling file, a detailed threshold analysis was performed in Excel by evaluating decision thresholds from 0.01 to 0.99. For each threshold, predicted class labels were generated, and true positives, false positives, true negatives and false negatives were calculated separately for training and test datasets. Based on these values, precision, recall, F1-score and confusion matrices were created for test and train data at 3 different thresholds and profitability were computed for all these 3 thresholds.
Model performance was first assessed using ROC curves for training and test data. The ROC curves showed strong separation between responders and non-responders, with similar shapes and AUC values for both datasets, indicating good generalization and no evidence of overfitting. Precision–Recall curves were also generated for training and testing data to evaluate performance under class imbalance, showing consistent and stable behavior across datasets. Threshold-wise precision and recall results for both training and test data were exported in MC_threshold_PR_train_test.csv. The optimal classification threshold was selected based on the highest F1 score on the test dataset, as test performance best reflects real-world deployment. The highest test F1-score was observed at a threshold of approximately 0.27, which provided the best balance between precision and recall. Importantly, this threshold also resulted in the highest profitability for both train and test, as calculated using the profit measures included in the modeling file. Therefore, threshold 0.27 was selected as the final operating point.
To further validate model stability, 5-fold cross-validation was performed on the training data. Predicted probabilities for each fold were exported to MC_cv_probs_5fold.csv. The cross-validated AUC was consistent with training and testing AUC values, confirming that the model's performance is robust and not sensitive to a single data split. Overall, the model demonstrates strong predictive performance, stable probability estimates, and an optimal threshold that maximizes both classification accuracy and business profitability.

4.3.K-Means (Financial Behavior)

To create a sophisticated segmentation that accounts for both customer value and retention risk, we constructed a K-Means model with **k=4 clusters**. The decision to use exactly four clusters was not arbitrary; it was determined using the **Elbow Method**. By plotting the "Within-Cluster Sum of Squares" (WSS) against the number of potential clusters, we identified that any number of clusters between 2 and 5 can be chosen. After creating all 4 models, it became clear that four segments provided the optimal balance, significantly reducing the variance within groups without over complicating the model with too many small and fragmented niches.
The clustering model used four key variables: **Income** (capacity), **Total Spend** (value), **Web Visits** (engagement) and **Recency** (churn risk). Before running the model, all variables were scaled so that variables with large values, such as Income (in the thousands), did not overpower variables with small values, like Web Visits (single digits), allowing the model to weight each behavioral trait equally.

Once the K-Means algorithm mathematically grouped the customers into these four optimal segments, we applied a **Hierarchical Decision Logic** to automatically label each group based on its dominant trait. The logic worked in a strict priority sequence:

I. **Stars (VIPs):** The code first identified the cluster with the highest average spending.
II. **At Risk:** From the remaining groups, it identified the cluster with the highest Recency (days since last purchase), isolating potential churners.
III. **Discount Hunters:** It then found the group with high web traffic but low spending, flagging them as bargain seekers.
IV. **Potentialists:** The final remaining group was characterized by high income and activity, was labeled as the growth opportunity.

This approach ensures that the segmentation is both statistically sound and easy to use for real business decisions.

4.4. K-Means (Spending Behavior)

We conducted a customer segmentation analysis using K-means clustering to group individuals based on their spending behavior across categories such as wines, fruits, meat, fish, sweets, and gold products. After scaling the data to ensure comparable weighting, using the elbow method, we chose 7 clusters. This segmentation was validated using silhouette scores to ensure cluster quality, and we concluded the analysis by profiling each group against their marketing response rates to identify the most engaged customer segments. Based on the cluster centers, we categorized every cluster as to what kinds of customers fall into each cluster, as shown in Table 1 in the appendix.

5. Policy Recommendation

5.1.K-Means

To better understand our customers in real life (not just in theory), I created a Cross Tabulation Matrix. This matrix shows where our **financial customer segments** overlap with our **product preference personas**. The boxes with the highest numbers (the "yellow cells") show the most

		k=4 clusters				
	Count of Cluster					
		1 (at risk)	2 (discount hunters)	3 (Potentialists)	4 (Stars)	Grand Total
MNT based custers	1 (sweet products, fish and gold)	2	8	58	49	117
	2(fish)	18	25	70	84	197
	3(low spenders)	581	565	15	46	1207
	4 (fruits, fish, sweet and meat)	2	3	70	70	145
	5 (wine)	19	35	76	108	238
	6 (Meat and Wine)			86	62	148
	7 (gold)	35	33	48	44	160
	Grand Total	657	669	423	463	2212

common customer types. These yellow cells help us understand the core makeup of our customers.

The strongest overlap we found was between **at risk customers and Discount Hunters** with **Low Spenders**. This confirms what our data models predicted. Customers who are financially low value also do not show strong interest in any specific product category. They mainly shop for low prices and do not have brand loyalty. This behavior aligns with the "Price Switcher" segment defined in standard marketing theory, where customers possess zero loyalty and are driven solely by price elasticity. According to studies by the Harvard Business Review, attempting to build loyalty with this segment often yields a negative Return on Investment (ROI), confirming our decision to minimize marketing spend on this group. (Reinartz, 2002)

On the other hand, the **Potentialist** group (customers with high income but medium spending) showed a strong connection with **Meat and Wine Enthusiasts**. This is an important insight. It tells us that wine and meat are the entry points for this group. They can be used to move these customers toward becoming VIPs by slowly increasing their spending.

We also found that our **"Stars" (VIP customers)** overlap strongly with **"Carnivores"** and **"Wine Enthusiasts."** This means that the quality of our **Fresh Meat** and **Premium Wine** selection is likely the primary attraction for our most profitable customers. For VIPs, these departments are not just products, they are the key drivers of their loyalty. This strategy is supported by (Zeithaml, 2001), who apply the Pareto Principle to the 'Customer Pyramid,' demonstrating that the top tier (Stars) generates the vast majority of profit. Furthermore, our data aligns with (Laurent, 1985) research on 'High-Involvement' products, confirming that loyalty in this top tier is anchored by critical categories like premium meat and wine rather than general transaction volume.

The real value of this analysis comes from combining two different models instead of using them separately. The first model grouped customers based on financial behavior. It told us who has money and how much they spend, but it did not tell us what they like. The second model showed product preferences, but it ignored how profitable the customer is.

To solve this, I created a **Dual Lens approach**, which looks at both financial behavior and product preference at the same time. This allows the company to stop pushing random products and instead pull customers in with offers that match both their taste and their spending ability. This makes marketing more relevant and more effective. This reflects the shift in modern CRM from 'Demographic Segmentation' to 'Psychographic and Behavioral Segmentation', which (Gallup) reports can outperform traditional methods by up to 85% in sales growth.

Based on this combined view, four clear policy directions were created.

I. For “**Potentialists”**, the goal is to increase how much they spend. These customers have the money to spend like VIPs but are currently buying only basic items, often from competitors. Giving them general discounts will not change their habits. Instead, we need bundles that encourage new behavior.

For Potentialists who like meat, we can offer a **Sunday Roast Bundle**. This includes premium meat with a complimentary bottle of red wine. This helps introduce them to wine and increases basket value. This strategy utilizes 'Market Basket Analysis' theory, supported by (Stremersch), whose research in the *Journal of Marketing* suggests that bundling a high frequency item (meat) with a lower-frequency impulse item (wine) significantly increases the overall basket size and profitability.
For Potentialists who prefer fish, we can offer a special deal where premium fresh salmon is priced the same as their usual frozen fish for the first purchase. This encourages them to upgrade and continue buying higher margin products in the future. This strategy relies on the 'Reference Dependence' model established by (Hardie, 1993) in *Marketing Science*. Their research confirms that once a consumer adopts a higher quality standard (the new reference point), a drop in quality is perceived as a psychological 'loss,' making them statistically unlikely to revert to the inferior option.

II. For "**VIP" customers**, the main focus is keeping them loyal, not offering discounts. The data shows that VIPs care more about value and experience than price. Sending them the same discounts as regular customers can actually harm the brand. This is supported by the (Institute, 2017) report *Loyalty Deciphered*, which found that while **97%** of loyalty programs rely on transactional rewards, high value customers are far more effectively retained through emotional drivers like exclusivity and access.
For VIPs who like wine, we should offer **early access** to new wine arrivals before anyone else. This makes them feel special without lowering prices.
For VIPs who enjoy sweets, we can add a free sample of a premium chocolate brand to their next order. This creates emotional loyalty and a feeling of appreciation that discounts cannot replace. This aligns with the 'Principle of Reciprocity' defined by (Cialdini, 2006) in his seminal text *Influence*, which posits that when a brand provides an unexpected value or gift, the recipient feels a subconscious psychological obligation to return the favor through continued loyalty.

III. The "**Discount Hunter"** group is useful mainly for managing inventory, even though they do not bring high revenue. These customers are very price sensitive, so we should use them to clear excess stock without showing big discounts to VIPs. This reflects the concept of 'Yield Management' as defined by (Kimes, 1989) in the *Journal of Operations Management*, which establishes that price-sensitive customers are valuable not for high profit margins, but for maximizing capacity utilization and inventory turnover.
If there is extra stock of sweets, we can send Discount Hunters who like sweets a bulk offer, such as "Buy 5, Get 50% Off." This helps move inventory quickly while keeping premium customers unaware of heavy discounts.
Because Low Spenders dislike shipping fees, we can slightly raise the free shipping threshold. For example, setting free shipping at $30 instead of $20 encourages them to add one more item to their cart, increasing average order value.

IV. For “**At Risk” customers**, standard reactivation emails do not work well. Messages like “We miss you” are too generic and usually get ignored. To bring these customers back, the message must remind them of something specific they liked. This strategy is validated by (Epsilon, 2018) research report *The Power of Me*, which found that **80%** of consumers are more likely to make a purchase when brands offer personalized experiences compared to generic interactions.
If an at Risk customer prefers fish, we should notify them when a product they previously bought, such as fresh tuna, is back in stock. This triggers memory and emotional connection.
For at Risk customers who buy luxury items like gold or accessories, messages about limited edition or new arrivals work better. These customers often respond to novelty and exclusivity rather than discounts. This tactic leverages the 'Scarcity Principle' as defined by (Cialdini, 2006), who demonstrates that customers assign higher value to items that are perceived as rare or rare in availability, often leading to immediate action more effectively than price incentives.

5.2.Decision Tree (Behavioral Filtering)

To move beyond broad segmentation and identify exactly *who* will accept our next offer, we constructed a Decision Tree model. Unlike the K-Means model which groups customers by similarity, the Decision Tree acts as a sequential filter, prioritizing customers based on the specific "signals" they send through their actions.

The model was built using the “rpart” algorithm in R, prioritizing interpretability over raw "black box" accuracy. It mathematically proved that customer behavior is not random; it follows a strict hierarchy. The most powerful predictor of future behavior is past campaign acceptance specifically Campaign 5, 3, and 1. If a customer has said "yes" before, they are exponentially more likely to say "yes" again. However, for those who said "no," the model switches tactics, looking for high spending (Total Spend/Meat) or digital engagement (Web Visits) to find hidden value.

By following the branches of the tree, we developed a "Sequential Targeting Logic" to maximize marketing ROI:

I. **The "Momentum" Strategy (Proven Responders)** This group includes anyone who accepted the most recent campaign (Campaign 5). The model identifies them immediately at the first split. They have a 54.9% probability of accepting again, compared to the 15% average. They are already "trained" to open our emails.
For new campaign we should not send them generic sales pitches. Instead, treat them as VIPs with "Early Access" offers. This aligns with the **"Endowed Progress Effect" (Nunes & Drèze, 2006)**, which suggests that people are more likely to complete an action if they feel they have already made progress toward it. Since these customers have already "started" the journey by buying once, framing the next offer as an exclusive next step rather than a new sales pitch capitalizes on their psychological momentum.

II. **The "Sleeping Giants" (High Value Recovery)** These customers rejected the last campaign, flagging them as "cold," but possess High Meat Spending (> 815.5). Their response probability jumps back up to 56.5% purely because of their purchasing power. The generic campaign failed because it lacked relevance. We need a "Product-Led Recovery" with a Premium Butcher's Bundle. Secondary research on **"Choice Architecture" (Thaler & Sunstein)** indicates that high-value buyers often suffer from decision fatigue; they reject generic "browse all" emails but respond well to curated, default options that match their known preferences (e.g., a pre-bundled meat selection).

III. **The "Digital Gold" Niche (Engaged Browsers)** These are long-tenure customers (> 2 years) who visit the website frequently and purchase Gold products. This yields a 76% response rate.
Use Digital Retargeting with "Flash Sale" pop-ups. This strategy leverages the **"Mere Exposure Effect" (Zajonc, 1968)**, which states that familiarity breeds preference. Since these customers are already exposing themselves to the brand frequently, a well-timed, low-friction prompt while they are active exploits their high "situational involvement" better than a cold email would.

IV. **The "Deal Hunters" (Low Engagement, High Deals)** These customers rarely visit the website but have high "Deals Purchases" (> 2.5). They have a 66% response rate, but only to specific triggers.
Send "Bulk Buy" offers (e.g., "Buy 3, Get 20% Off"). Research on **"Deal Proneness" (Lichtenstein et al., 1990)** defines this segment not as brand loyalists, but as "transaction utility" seekers—they derive pleasure specifically from the *saving* itself, not just the product. Marketing to them must emphasize the mathematical "win" of the discount rather than brand values.

V. **The "Do Not Disturb" List (Efficiency Gains)** Customers who rejected past campaigns, have low spending, and short tenure (Response rate: ~6%).
Suppress marketing spend. According to **Schultz's "Customer Equity" framework**, investing in negative-ROI segments destroys shareholder value. Furthermore, excessive emailing to disinterested users triggers "list fatigue," damaging domain reputation and deliverability for the high-value segments.

5.3.Logistic Regression (Precision Scoring)

While K-Means grouped customers and the Decision Tree filtered them, Logistic Regression ranks every customer with a precise probability. The model achieved an AUC of 0.87, driven by Momentum, Urgency (Recency), and Specific Tastes.

I. **The "Momentum" Policy (The 60x Multiplier)** The single most powerful predictor is AcceptedCmp5. A customer who accepted the last campaign is 60.1 times (Odds Ratio) more likely to accept the next one.

Implement a "Zero-Friction Upsell." Do not distract these users with awareness campaigns. This validates **Cialdini's Principle of Consistency (1984)**, which argues that once an individual makes a choice (buying Campaign 5), they encounter personal and interpersonal pressure to behave consistently with that commitment. The marketing message should be "Since you enjoyed X, you deserve Y," leveraging their internal desire for consistency.

II. **The "PhD Privilege" Policy (Targeting Intellect)** Customers with a PhD are 1.83 times more likely to respond than those with basic education.
Use "Connoisseur" Messaging focusing on specs and origin. This demographic exhibits a high **"Need for Cognition" (Cacioppo & Petty, 1982)**. Individuals with high Need for Cognition prefer complex, verbal, and information-rich messages over peripheral cues like celebrity endorsements or flashy colors. Our copy must respect their intellect to convert them.

III. **The "Recency Decay" Rule (The Silent Killer)** Recency has an odds ratio of 0.96; for every day without a purchase, probability drops by ~4%.
Implement a "30-Day Cliff" reactivation trigger. This is grounded in the **"Forgetting Curve" (Ebbinghaus)** applied to marketing: brand recall decays exponentially over time. Intervening at Day 25 creates a "memory cue" before the brand falls below the threshold of active recall, which is significantly cheaper than trying to re-acquire a churned customer later (The 5x Rule of Acquisition Cost).

IV. **The "Discount Addiction" Trap (Managing Margins)** NumDealsPurchases is a positive predictor (Odds Ratio 1.25), but these customers hurt margins.
Create a "Secret Clearance" segment. This aligns with **"Price Discrimination" theory (Pigou)**, which suggests charging different prices to different segments based on their willingness to pay. We should extract volume from this price-sensitive group using clearance stock, while shielding our full-price inventory from them to prevent "reference price erosion" among our premium buyers.

6. Way Forward

While this project successfully identified key customer segments and predictive factors using K-Means, Decision Trees, and Logistic Regression, several advanced analytical methods could further enhance the precision of our marketing strategies given more time and resources.

6.1.Advanced Predictive Modeling

In this analysis, we utilized a single Decision Tree primarily for its interpretability and to understand the sequence of customer decisions. However, the single tree showed lower precision on the test set (PR-AUC = 0.442). To improve predictive accuracy and reduce variance, we would implement **Random Forest** or **Gradient Boosting (XGBoost)** models. These ensemble

methods would aggregate predictions from multiple trees to better handle the class imbalance and improve the identification of "Responders" versus "Non-Responders".

6.2. Market Basket Analysis

Our current recommendation strategy relies on a manual "Cross Tabulation Matrix" and a "Dual Lens" approach to identify product overlaps, such as the connection between "Potentialists" and "Meat and Wine" products. With more time, we would employ the **Apriori Algorithm** or **Association Rule Mining**. This would statistically quantify the strength of links between product categories (e.g., if a customer buys Meat, they are 70% likely to buy Wine), allowing for automated, data-driven bundle creation rather than manual inference.

6.3. Customer Lifetime Value (CLV) Prediction

The current analysis segments customers based on historical financial behavior (Income, Total Spend) and Recency. A valuable next step would be building a **CLV Regression Model**. Instead of just looking at past spending, this model would predict the *future* monetary value of each customer segment over the next 12–24 months. This would allow the company to allocate higher marketing budgets to "Potentialists" who show high future growth probability, rather than just high current income.

6.4. Dimensionality Reduction (PCA)

For our K-Means clustering, we manually selected variables such as Income, Total Spend, and Web Visits. To capture more complex behavioral patterns without overfitting, we would apply **Principal Component Analysis (PCA)**. PCA would allow us to condense the 29 available features into principal components, potentially revealing hidden customer traits that are currently obscured by correlations between variables like MntWines and MntMeatProducts.

6.5. A/B Testing Framework

Finally, the "Policy Recommendations" section suggests specific tactics, such as offering "Sunday Roast Bundles" to Potentialists or "Early Access" to VIPs. The logical next step is to design a formal **A/B Testing Framework**. This would involve deploying these offers to small control and treatment groups within the identified clusters to empirically validate the response rates before a full-scale rollout.

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Appendix References

Appendix A: Detailed R summary and Descriptive Statistics

```
> # 13. Summary statistics
> summary(final_data)
      ID          Year_Birth       Education    Marital_Status         Income          Kidhome          Teenhome
 Min.   :    0   Min.   :1893   2n Cycle  : 200   Length:2215        Min.   :  1730   Min.   :0.0000   Min.   :0.0000
 1st Qu.: 2814   1st Qu.:1959   Basic     :  54   Class :character   1st Qu.: 35284   1st Qu.:0.0000   1st Qu.:0.0000
 Median : 5455   Median :1970   Graduation:1115   Mode  :character   Median : 51373   Median :0.0000   Median :0.0000
 Mean   : 5587   Mean   :1969   Master    : 365                      Mean   : 51970   Mean   :0.4415   Mean   :0.5056
 3rd Qu.: 8419   3rd Qu.:1977   PhD       : 481                      3rd Qu.: 68487   3rd Qu.:1.0000   3rd Qu.:1.0000
 Max.   :11191   Max.   :1996                                        Max.   :162397   Max.   :2.0000   Max.   :2.0000
  Dt_Customer          Recency         MntWines        MntFruits      MntMeatProducts  MntFishProducts  MntSweetProducts
 Min.   :2012-07-30   Min.   : 0.00   Min.   :   0.0   Min.   :  0.00   Min.   :   0.0   Min.   :  0.00   Min.   :  0.00
 1st Qu.:2013-01-16   1st Qu.:24.00   1st Qu.:  24.0   1st Qu.:  2.00   1st Qu.:  16.0   1st Qu.:  3.00   1st Qu.:  1.00
 Median :2013-07-09   Median :49.00   Median : 175.0   Median :  8.00   Median :  68.0   Median : 12.00   Median :  8.00
 Mean   :2013-07-10   Mean   :49.02   Mean   : 305.2   Mean   : 26.36   Mean   : 167.1   Mean   : 37.65   Mean   : 27.04
 3rd Qu.:2013-12-31   3rd Qu.:74.00   3rd Qu.: 505.0   3rd Qu.: 33.00   3rd Qu.: 232.5   3rd Qu.: 50.00   3rd Qu.: 33.00
 Max.   :2014-06-29   Max.   :99.00   Max.   :1493.0   Max.   :199.00   Max.   :1725.0   Max.   :259.00   Max.   :262.00
  MntGoldProds    NumDealsPurchases NumWebPurchases  NumCatalogPurchases NumStorePurchases NumWebVisitsMonth
 Min.   :  0.00   Min.   : 0.000    Min.   : 0.000   Min.   : 0.000      Min.   : 0.000    Min.   : 0.000
 1st Qu.:  9.00   1st Qu.: 1.000    1st Qu.: 2.000   1st Qu.: 0.000      1st Qu.: 3.000    1st Qu.: 3.000
 Median : 25.00   Median : 2.000    Median : 4.000   Median : 2.000      Median : 5.000    Median : 6.000
 Mean   : 43.98   Mean   : 2.323    Mean   : 4.086   Mean   : 2.672      Mean   : 5.802    Mean   : 5.319
 3rd Qu.: 56.00   3rd Qu.: 3.000    3rd Qu.: 6.000   3rd Qu.: 4.000      3rd Qu.: 8.000    3rd Qu.: 7.000
 Max.   :321.00   Max.   :15.000    Max.   :27.000   Max.   :28.000      Max.   :13.000    Max.   :20.000
  AcceptedCmp3     AcceptedCmp4      AcceptedCmp5      AcceptedCmp1      AcceptedCmp2     Complain Response
 Min.   :0.00000   Min.   :0.00000   Min.   :0.00000   Min.   :0.00000   Min.   :0.00000  0:2194   0:1882
 1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.00000  1:  21   1: 333
 Median :0.00000   Median :0.00000   Median :0.00000   Median :0.00000   Median :0.00000
 Mean   :0.07359   Mean   :0.07404   Mean   :0.07314   Mean   :0.06411   Mean   :0.01354
 3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.:0.00000
 Max.   :1.00000   Max.   :1.00000   Max.   :1.00000   Max.   :1.00000   Max.   :1.00000
```

```
> skim(final_data)
── Data Summary ────────────────────────
                           Values
Name                       final_data
Number of rows             2215
Number of columns          27
_______________________
Column type frequency:
  character                1
  Date                     1
  factor                   3
  numeric                  22
_______________________
Group variables            None

── Variable type: character ────────────────────────────────────────────────────────────────────────────────
  skin_variable  n_missing complete_rate min max empty n_unique whitespace
1 Marital_Status         0             1   4   8     0        8          0

── Variable type: Date ─────────────────────────────────────────────────────────────────────────────────────
  skin_variable n_missing complete_rate min         max        median     n_unique
1 Dt_Customer           0             1 2012-07-30 2014-06-29 2013-07-09      662

── Variable type: factor ───────────────────────────────────────────────────────────────────────────────────
  skin_variable n_missing complete_rate ordered n_unique top_counts
1 Education             0             1 FALSE         5 Gra: 1115, PhD: 481, Mas: 365, 2n : 200
2 Complain              0             1 FALSE         2 0: 2194, 1: 21
3 Response              0             1 FALSE         2 0: 1882, 1: 333

── Variable type: numeric ──────────────────────────────────────────────────────────────────────────────────
  skim_variable       n_missing complete_rate       mean        sd   p0    p25   p50    p75   p100 hist
1 ID                          0             1  5587.     3249.      0  2814.  5455  8419   11191 ▇▇▇▇▇
2 Year_Birth                  0             1  1969.       12.   1893  1959  1970  1977    1996 ▁▁▂▅▇
3 Income                      0             1 51970.    21526.   1730 35284 51373 68487  162397 ▂▇▅▁▁
4 Kidhome                     0             1     0.442    0.537    0     0     0     1       2 ▇▅▁▂▁
5 Teenhome                    0             1     0.506    0.544    0     0     0     1       2 ▇▅▁▂▁
6 Recency                     0             1    49.0     28.9      0    24    49    74      99 ▇▇▇▇▇
7 MntWines                    0             1   305.     337.       0    24   175   505    1493 ▇▂▁▁▁
8 MntFruits                   0             1    26.4     39.8      0     2     8    33     199 ▇▁▁▁▁
9 MntMeatProducts             0             1   167.     224.       0    16    68   232.   1725 ▇▁▁▁▁
10 MntFishProducts            0             1    37.7     54.8      0     3    12    50     259 ▇▁▁▁▁
11 MntSweetProducts           0             1    27.0     41.1      0     1     8    33     262 ▇▁▁▁▁
12 MntGoldProds               0             1    44.0     51.8      0     9    25    56     321 ▇▁▁▁▁
13 NumDealsPurchases          0             1     2.32     1.92     0     1     2     3      15 ▇▁▁▁▁
14 NumWebPurchases            0             1     4.09     2.74     0     2     4     6      27 ▇▁▁▁▁
15 NumCatalogPurchases        0             1     2.67     2.93     0     0     2     4      28 ▇▁▁▁▁
16 NumStorePurchases          0             1     5.80     3.25     0     3     5     8      13 ▇▅▅▅▅
17 NumWebVisitsMonth          0             1     5.32     2.43     0     3     6     7      20 ▅▇▅▅▅
18 AcceptedCmp3               0             1     0.0736   0.261    0     0     0     0       1 ▇▁▁▁▁
19 AcceptedCmp4               0             1     0.0740   0.262    0     0     0     0       1 ▇▁▁▁▁
20 AcceptedCmp5               0             1     0.0731   0.260    0     0     0     0       1 ▇▁▁▁▁
21 AcceptedCmp1               0             1     0.0641   0.245    0     0     0     0       1 ▇▁▁▁▁
22 AcceptedCmp2               0             1     0.0135   0.116    0     0     0     0       1 ▇▁▁▁▁
```

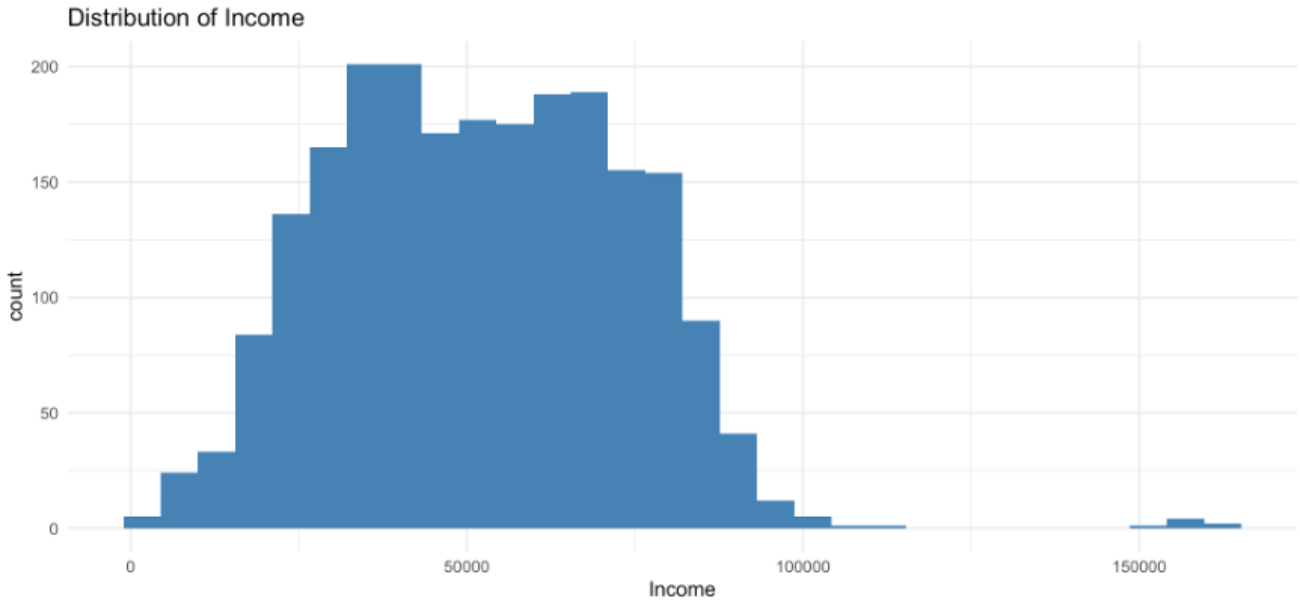
Appendix B: Additional EDA visualizations (all plots)

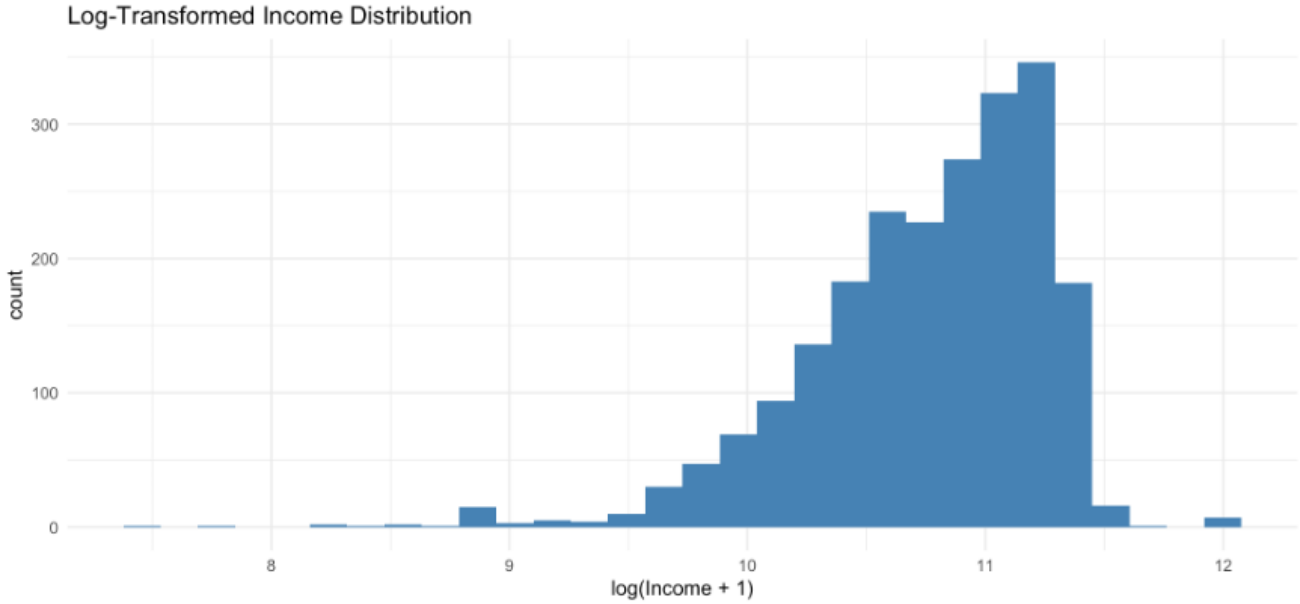
B.1 Response Variable Distribution (Dependent variable)

```
> # 14. Check Response class balance
> table(final_data$Response)

   0    1
1882  333
> prop.table(table(final_data$Response))

         0         1
0.8496614 0.1503386
```

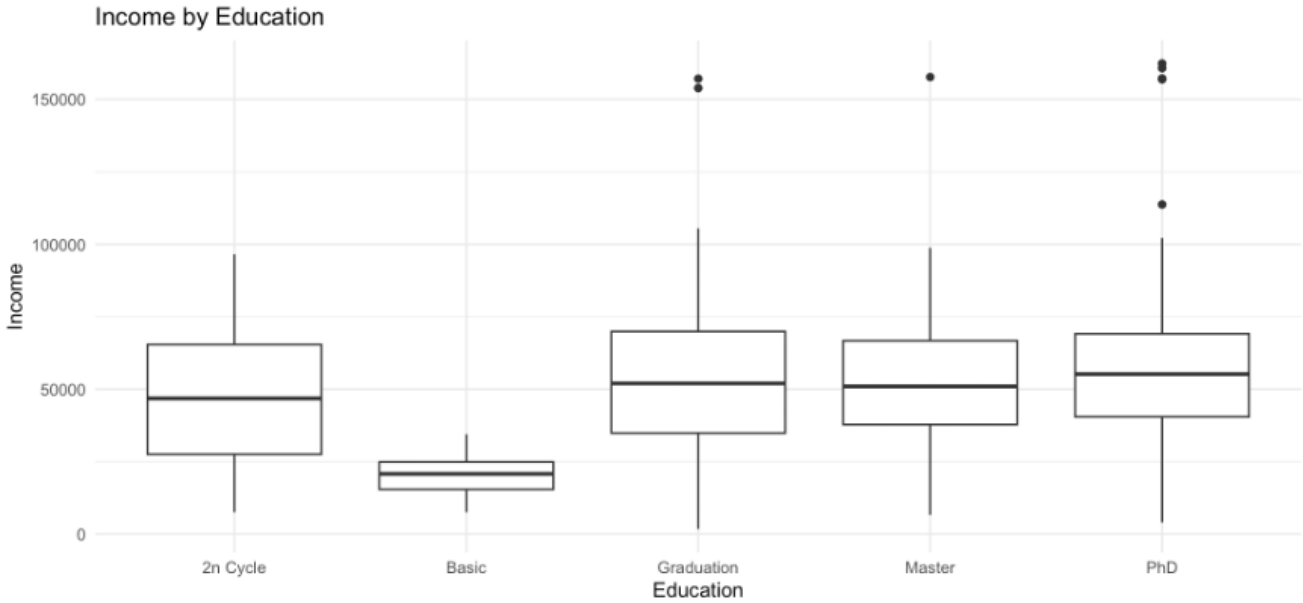
Response Status	Count	Percentage
No Response (0)	1,882	84.97%
Yes Response (1)	333	15.03%

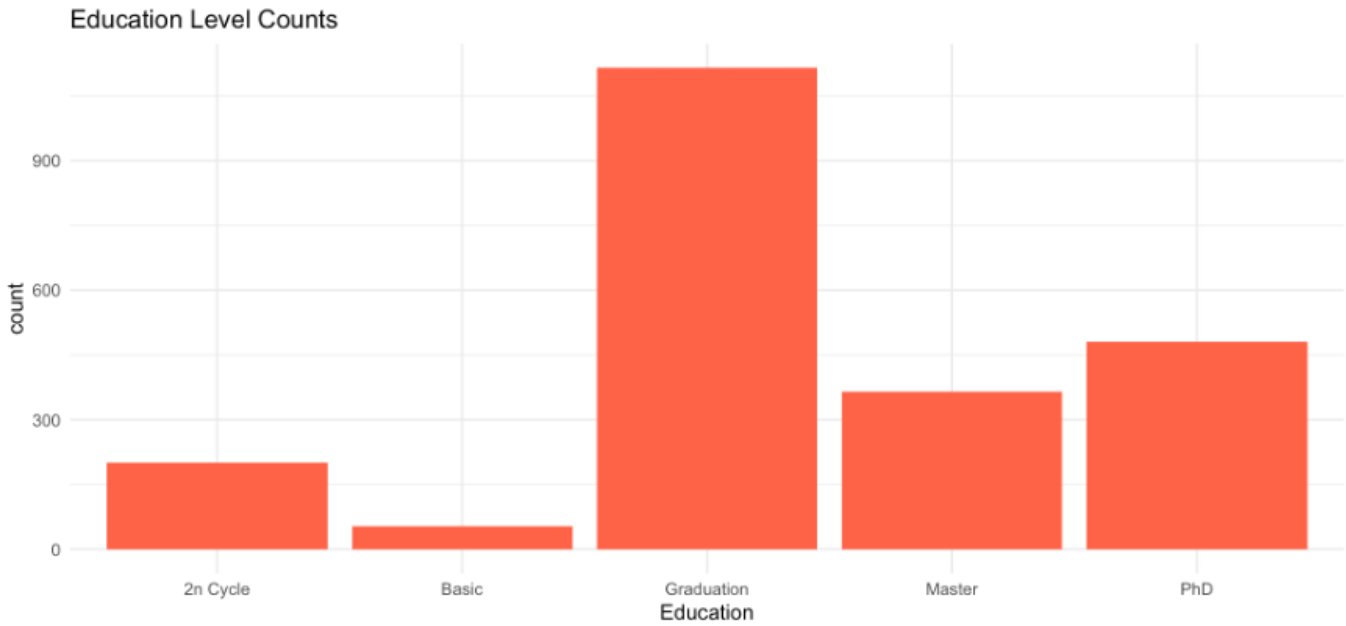
B.2 Income Distribution and Transformation



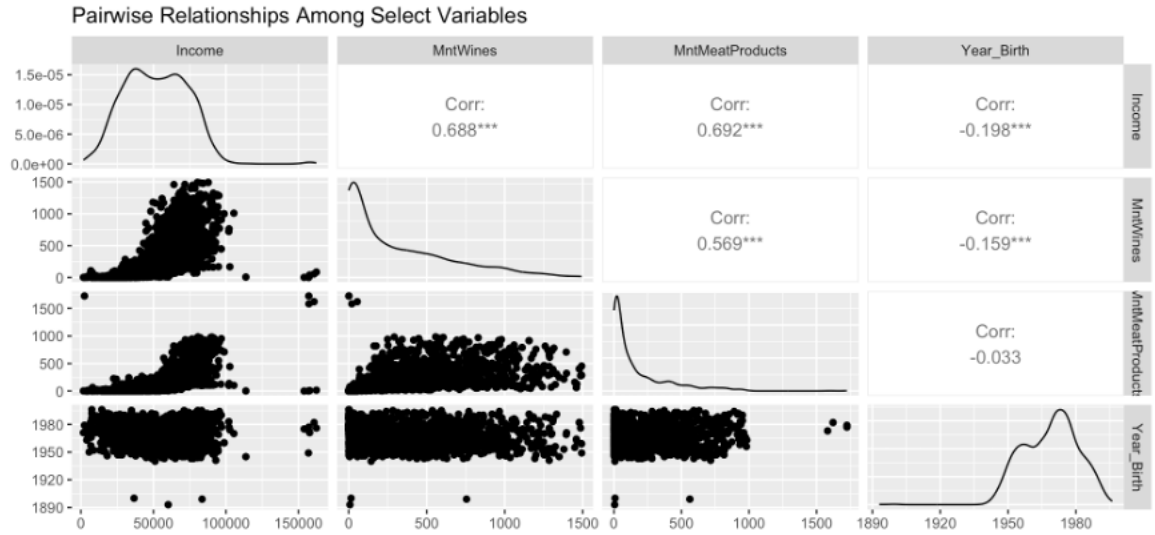


B.3 Relationship Between Education and Income

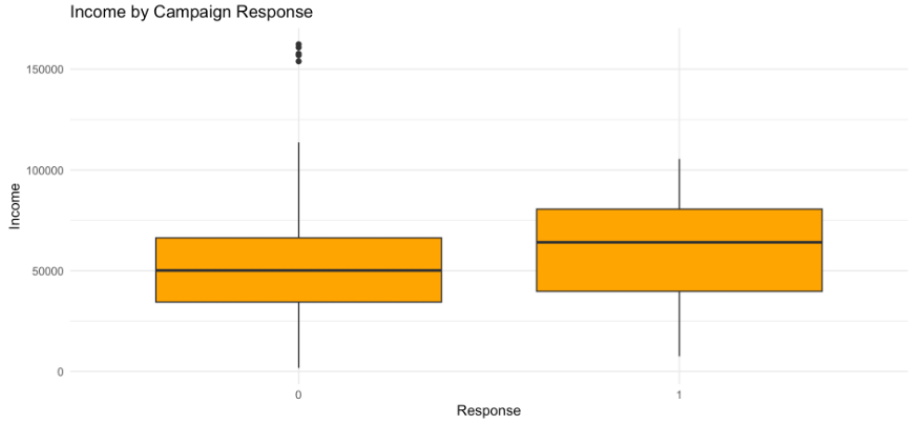




B.4 Income and Product Spending



B.5 Income and Campaign Response

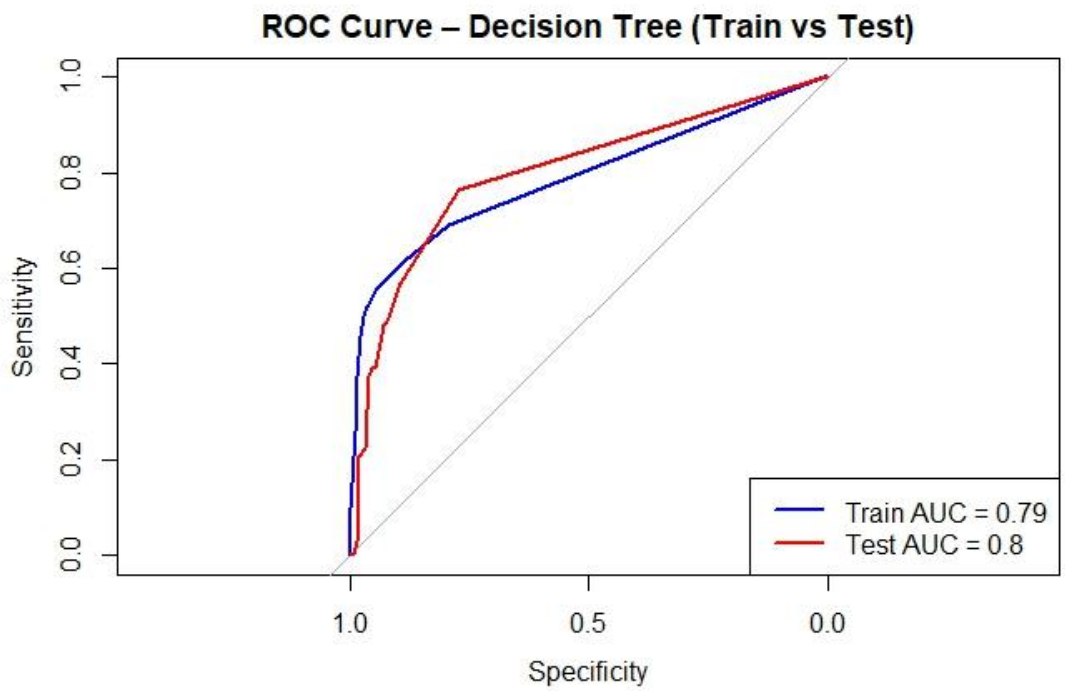


Appendix C: Decision Tree Model

C.1 Pruned Decision Tree

```
n= 1548

node), split, n, loss, yval, (yprob)
      * denotes terminal node

  1) root 1548 231 0 (0.85077519 0.14922481)
    2) AcceptedCmp5< 0.5 1435 169 0 (0.88222997 0.11777003)
      4) AcceptedCmp3< 0.5 1340 132 0 (0.90149254 0.09850746)
        8) MntMeatProducts< 815.5 1317 119 0 (0.90964313 0.09035687)
         16) Tenure_Days< 746.5 1107  71 0 (0.93586269 0.06413731) *
         17) Tenure_Days>=746.5 210  48 0 (0.77142857 0.22857143)
           34) NumWebVisitsMonth< 8.5 185  35 0 (0.81081081 0.18918919)
             68) Education=2n Cycle,Basic,Graduation,Master 144  18 0 (0.87500000 0.12500000) *
             69) Education=PhD 41  17 0 (0.58536585 0.41463415)
              138) NumDealsPurchases< 2.5 20   3 0 (0.85000000 0.15000000) *
              139) NumDealsPurchases>=2.5 21   7 1 (0.33333333 0.66666667)
                278) MntSweetProducts< 0.5 7   2 0 (0.71428571 0.28571429) *
                279) MntSweetProducts>=0.5 14   2 1 (0.14285714 0.85714286) *
           35) NumWebVisitsMonth>=8.5 25  12 1 (0.48000000 0.52000000)
             70) MntGoldProds< 9.5 8   0 0 (1.00000000 0.00000000) *
             71) MntGoldProds>=9.5 17   4 1 (0.23529412 0.76470588) *
        9) MntMeatProducts>=815.5 23  10 1 (0.43478261 0.56521739)
         18) Year_Birth>=1972.5 11   2 0 (0.81818182 0.18181818) *
         19) Year_Birth< 1972.5 12   1 1 (0.08333333 0.91666667) *
      5) AcceptedCmp3>=0.5 95  37 0 (0.61052632 0.38947368)
       10) Tenure_Days< 668 66  17 0 (0.74242424 0.25757576)
         20) Recency>=19.5 52   8 0 (0.84615385 0.15384615) *
         21) Recency< 19.5 14   5 1 (0.35714286 0.64285714) *
       11) Tenure_Days>=668 29   9 1 (0.31034483 0.68965517) *
    3) AcceptedCmp5>=0.5 113  51 1 (0.45132743 0.54867257)
      6) Tenure_Days< 396 39   9 0 (0.76923077 0.23076923) *
      7) Tenure_Days>=396 74  21 1 (0.28378378 0.71621622)
       14) Recency>=84.5 9   3 0 (0.66666667 0.33333333) *
       15) Recency< 84.5 65  15 1 (0.23076923 0.76923077) *
```

C.2 ROC Curve



Appendix D: Logistic Regression

Model Output and Probability Files

D.1 Probability and Profitability File

MC_modellingfile.csv

This file contains the predicted probabilities of output of the logistic regression model for both training and test datasets. Each row represents an individual customer and includes the following information:

Customer ID

Actual response (0 = No, 1 = Yes)

Dataset indicator (Train or Test)

Predicted probability of response

Profit values associated with each prediction

Using this file, the decision variable was created using predicted probabilities and assumed threshold of 0.4 and using that variable confusion matrix was constructed then across multiple thresholds ranging from 0.01 to 0.99, a table was created which include TN, PN,FN, FP, recall, precision, and F1 score of all these thresholds. For the highest F1 score of train and test data, confusion matrices were constructed, and profitability was calculated based on correct and incorrect classifications. This file served as the primary input for threshold optimization and profit maximization analysis. The highest profitability for the test and train dataset was observed at a threshold of approximately **0.27**, which also corresponded to the highest test F1 score. This confirms that the selected threshold balances both predictive performance and business value.

D.2 Precision & Recall Table

MC_threshold_PR_train_test.csv –

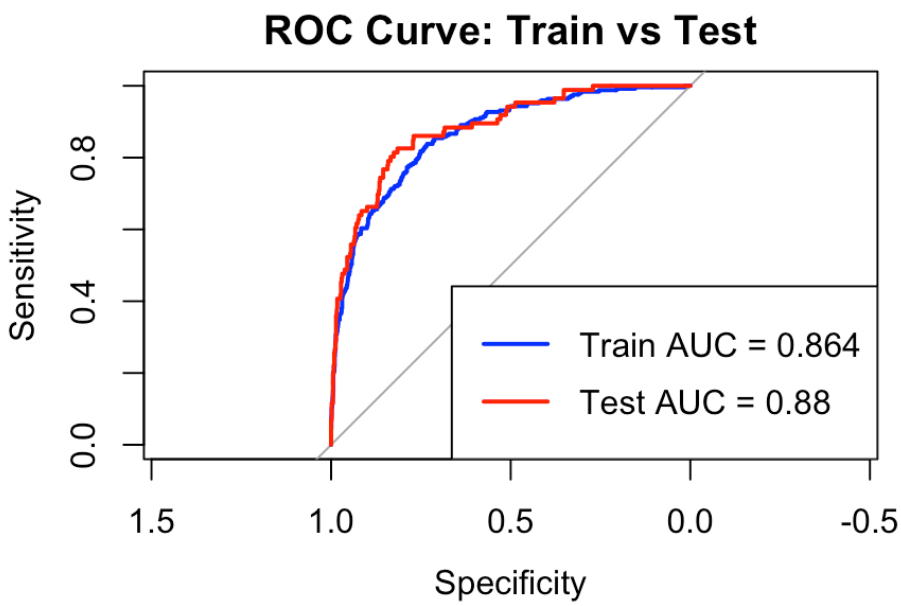
This file summarizes threshold-wise performance and contains:

- Threshold values (0.01–0.99)
- Precision (Train & Test)
- Recall (Train & Test)

The table illustrates how increasing the threshold improves precision but reduces recall, while lower thresholds increase recall at the cost of precision. This trade-off was central in identifying the optimal threshold.

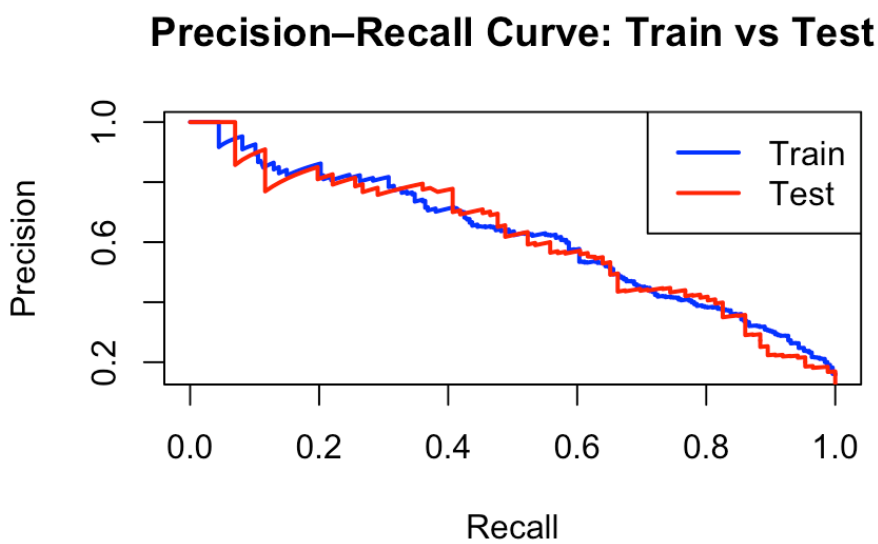
D.3 ROC Curve (Train vs Test)

The ROC curve plots the True Positive Rate against the False Positive Rate across all thresholds. Separate curves were plotted for training and test data (in an excel file MC_modellingfile.csv). Both curves lie well above the diagonal reference line, indicating strong discrimination between responders and non-responders. The similarity between the train and test curves suggests good generalization and no overfitting.



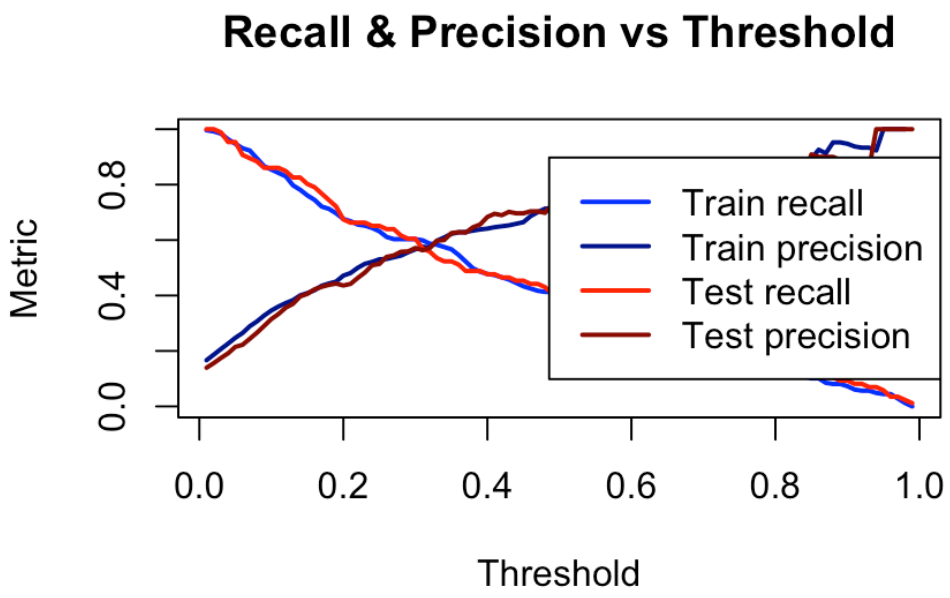
D.4 Precision–Recall Curve (Train vs Test)

The Precision–Recall curve evaluates model performance under class imbalance. The close alignment of train and test curves indicates consistent performance and reliability in identifying true responders without excessive false positives.



D.5 Threshold vs Precision and Recall Graph

This graph shows how precision and recalls change as the classification threshold varies from 0.01 to 0.99. Precision increases with higher thresholds, while recall decreases, clearly visualizing the trade-off involved in threshold selection. The selected threshold (≈ 0.27) lies in a region where both metrics are balanced.



D.6 5-Fold Cross-Validation Probabilities

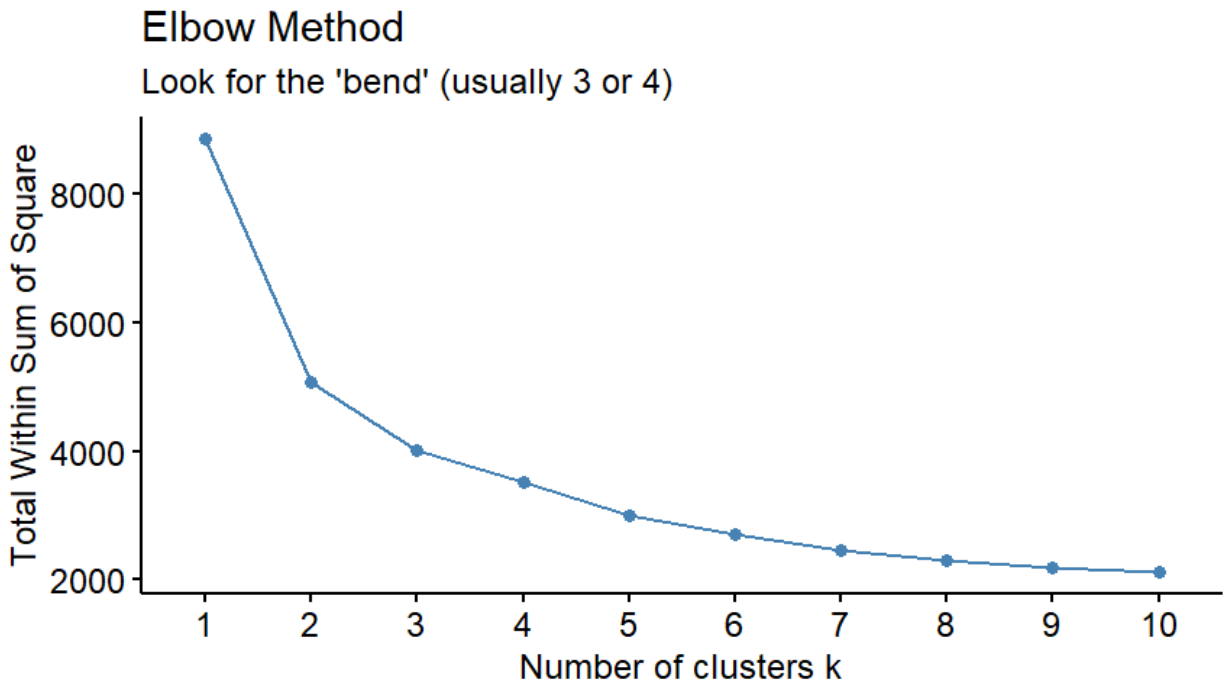
MC_cv_probs_5fold.csv

This file contains predicted probabilities from each fold of the 5-fold cross-validation procedure, including:

- Fold identifier
- Observed class
- Predicted probability

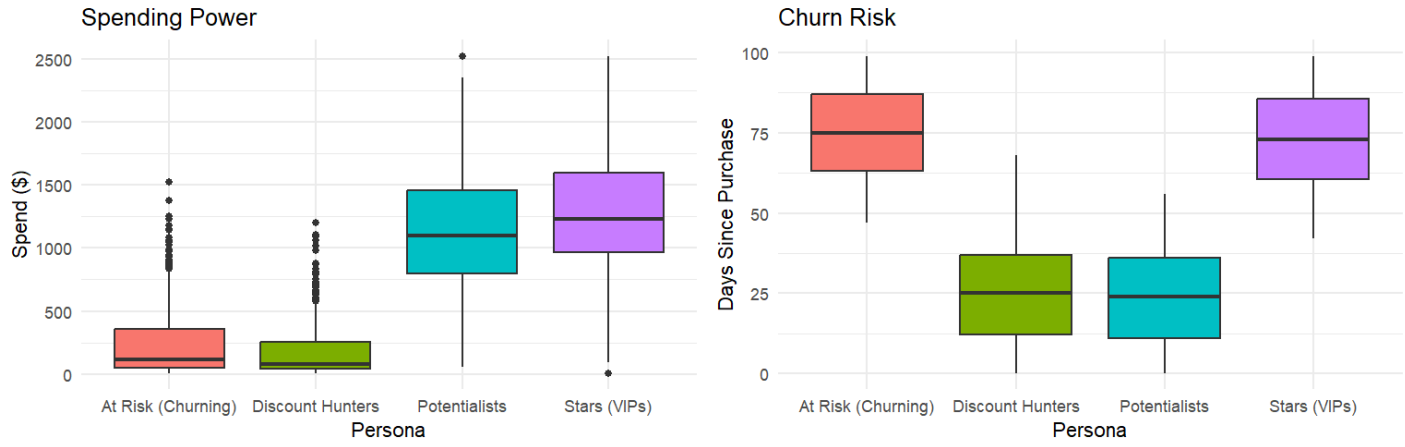
These results demonstrate that model performance is consistent across different subsets of the data. The cross-validated AUC closely matches the training and test of AUC values, confirming that the model's performance is stable and not dependent on a single train–test split.

Appendix E: Policy Recommendations



	Persona	Count	Avg_Income	Avg_Spend	Avg_Visits	Avg_Recency	Conversion_Rate
	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	At Risk (Churning)	669	39946	239	6.4	75	5.2
2	Discount Hunters	657	35800	177	6.8	25	16.7
3	Potentialists	463	71805	1134	3.7	24	25.5
4	Stars (VIPs)	423	74332	1282	3.1	73	16.5

summary of all 4 clusters for financial behavior



Visualization of persona for financial behavior

Cluster	Key Characteristic (Highest Z-Scores)	Interpretation / Persona
1	**Sweet Products (2.83)**, Fish (1.22), Gold (1.05)	**The Sweet Tooths:** They have a massive preference for sweets (almost 3 standard deviations above average). They also like fish and gold.
2	**Fish Products (1.07)**	**Fish Lovers:** Their only standout trait is buying fish. They are average or slightly below average on everything else.
3	**Everything Negative**	**Low Spenders / Budget:** Every single value is negative. These customers spend significantly less than average on *everything*.
4	**Fruits (2.79)**, Fish (1.81), Sweets (1.32), Meat (1.25)	**The Fresh Food Whales:** These are high-value customers who buy huge amounts of fresh food (Fruits, Fish, Meat). Note that their Wine score (0.65) is positive but not their defining feature.
5	**Wines (1.43)**	**Wine Enthusiasts:** They spend a lot on Wine, but they are **negative** (below average) on Fruits, Fish, Sweets, and Gold. They come specifically for the alcohol.
6	**Meat (2.32)**, Wines (1.29)	**The Carnivores / Gourmet:** They are defined by massive meat consumption, paired with high wine spending.
7	**Gold (2.15)**	**Gold / Asset Buyers:** Their gold spending is huge (2.15). Everything else is near zero (average) or slightly positive/negative.

Contribution Statements

Hamza Memon	26145	Conducted initial EDA then made the model of k means clustering based on income, total spend, website visits and recency and then cross tabulated both the Kmeans clusters. Worked on the policy recommendation part for K means clustering.
Sanjana kumari	30177	Loaded and inspected the dataset, performed data cleaning, detected outliers, and created visualizations to uncover key customer behavior patterns and predictors. Built logistic regression models using ROC AUC evaluation, cross-validation (CV), confusion matrices, and threshold optimization for reliable predictive performance. Worked on the report sections: problem statement outlining project objectives, data dictionary explaining variables, EDA findings summary, and model-building methodology.
Rabeen Adnan	30303	Performed exploratory data analysis to uncover trends and patterns in customer behavior. Built a decision tree model to predict campaign response and highlight the key variables influencing customer decisions. Concluded the project by handling the report formatting
Haseeb Ahmed	26062	Made the model for K-Means based on "NumDealsPurchases", "NumWebPurchases", "NumCatalogPurchases", "NumStorePurchases", "NumWebVisitsMonth". But the model was not as effective as model with "Mnt" columns, so it was not implemented. Worked on the Policy Recommendation for Decision Tree and Logistic Regression.
Hitesh Kumar	26652	Conducted initial EDA and then Made the model for k means clustering based on Mnt spending columns. Worked on the "Way Forward" part in the report.